Arduino Microcontroller Concepts

Chapter 5B • Simple Digital and Analog Input

ANALOG INPUT

Program: B. TECH., 7th Semester

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Learning goals

- Relate a variable input voltage to the numerical value returned by analogRead().
- Understand the 10-bit ADC idea and the relationship between voltage and ADC code.
- Convert ADC readings into voltage using the correct reference voltage.
- Map ADC values into useful application ranges with map().
- Use a 4051 multiplexer when more analog channels are required.
- Create threshold responses and safely measure higher voltages using a divider.

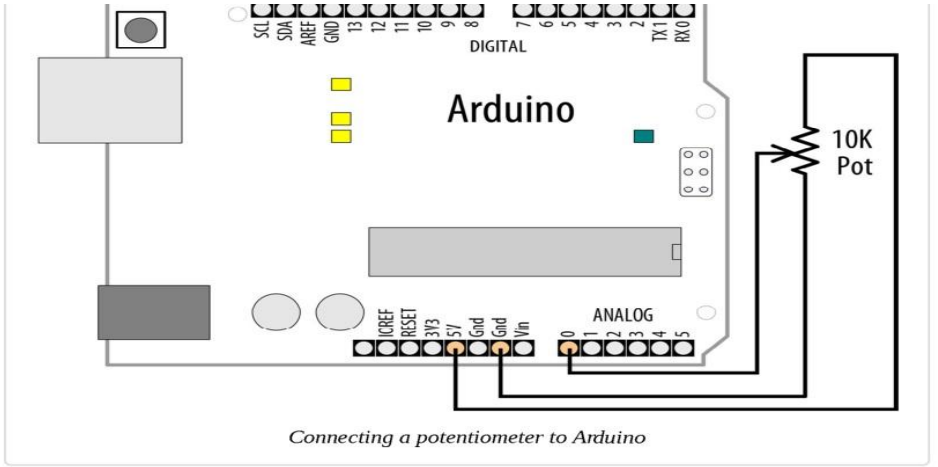
What is an analog input?

- An analog input can vary continuously over a voltage range.
- A potentiometer is a simple electrical component that can provide a variable voltage.
- A sensor may similarly produce a variable voltage related to a physical quantity.
- But the Arduino does not store the voltage directly: the ADC converts it into a digital number.

Potentiometer: the component behind many examples

Three terminals

- Two ends connect across the supply rails.
- The middle terminal is the wiper.
- Rotating the shaft changes the wiper voltage.



Connecting a potentiometer to Arduino

Discussion

This sketch uses the **analogRead** function to read the voltage on the potentiometer's *wiper* (the center pin). A pot has three pins; two are connected to a resistive material and the third pin (usually in the middle) is connected to a wiper that can be rotated to make contact anywhere on the resistive material. As the potentiometer rotates, the resistance between the wiper and one of the pins increases, while the other decreases. The

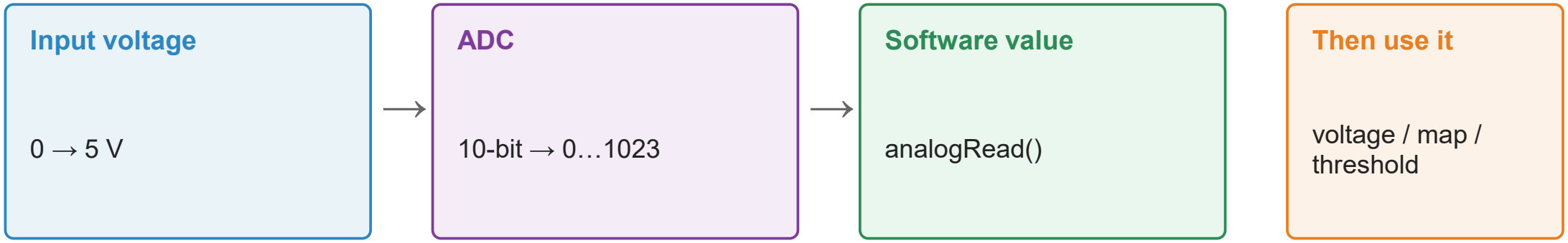
How it becomes an input

The wiper voltage is applied to an analog pin; as the wiper moves, the analog voltage changes and becomes an ADC code.

Potentiometer as a voltage divider

- The potentiometer is a variable resistive divider.
- With one end at 5 V and the other at ground, the wiper can provide a voltage between approximately 0 and 5 V.
- Moving the wiper changes the resistance ratio and therefore the output voltage.
- This is why a simple 10 kΩ potentiometer is useful for demonstrations and calibration.

ADC: from voltage to a number



The essential mental model: physical voltage → discrete code → application logic.

ADC calculation: input voltage ↔ ADC code

$$ADC \approx Vin / Vref \times 1023$$

$$Vin \approx ADC / 1023 \times Vref$$

Example 1

Vin = 2.5 V
Vref = 5 V
ADC ≈ 512

Example 2

ADC = 614
Vref = 5 V
Vin ≈ 3.00 V

Resolution

One count ≈ 5/1023 ≈ 4.89 mV

Reference voltage matters

- The ADC code represents a fraction of the reference voltage.
- If Vref changes, the same ADC code corresponds to a different input voltage.
- Always identify the correct reference before solving a measurement problem.
- For ADC = 512: Vref = 5 V gives ≈ 2.50 V; Vref = 4.096 V gives ≈ 2.05 V.

Reading analog values in Arduino


```
int analogPin = A0;
int val = 0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  val = analogRead(analogPin);
  Serial.println(val);
}
```

analogRead(pin)

Returns a numerical ADC code.

Serial Monitor

Useful for inspecting sensor values during development.

Exam skill

Given an ADC code, calculate or infer the voltage.

Changing the range with map()

map(x, 0, 1023, 0, 255)

Input

0 → 1023

Output

0 → 255

Example

512 → about 128

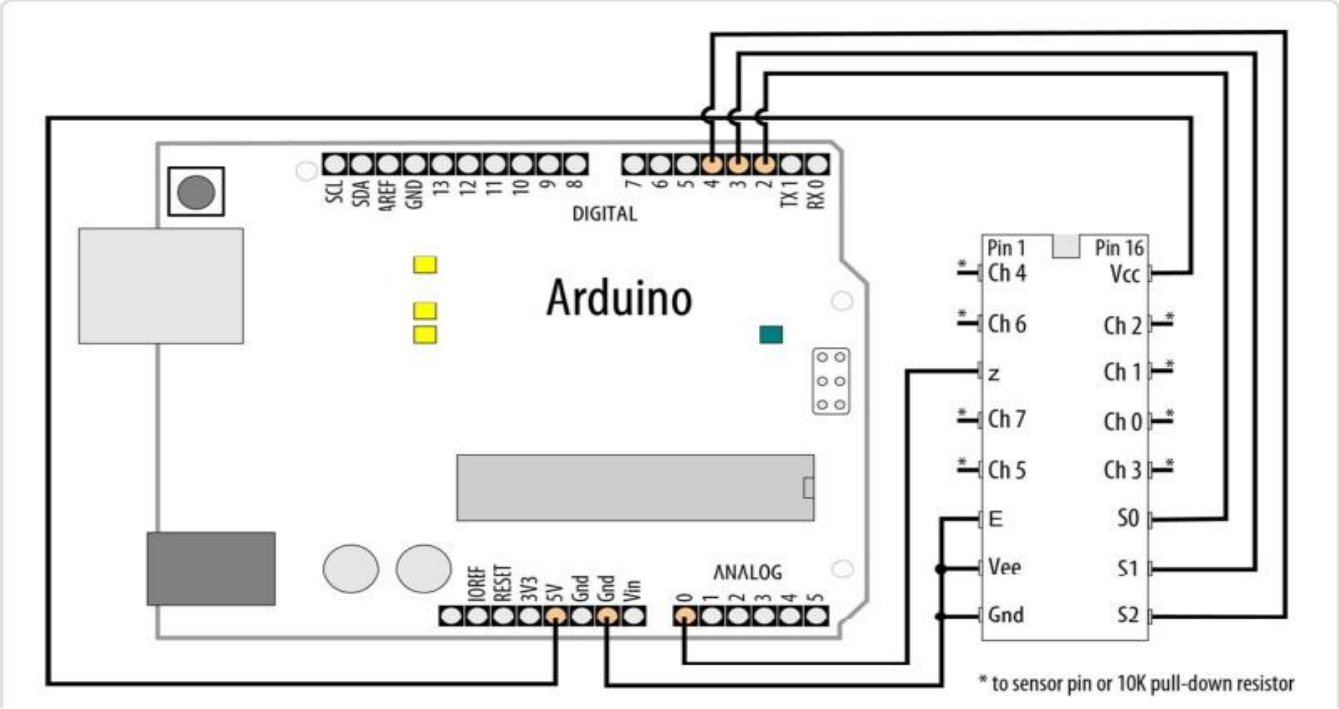
Important: map() changes the numerical scale. It does not itself create the physical output.

Why map() is useful in applications

- A raw ADC code may not match the range required by the application.
- You can map 0–1023 to 0–100 for percentage, 0–180 for angle, or 0–255 for a PWM control value.
- The mapped number can then be used to drive an output or make a decision.
- Mapping is a software transformation, not a physical voltage change.

More than six analog inputs: 4051 multiplexer


```
        Serial.print("Channel ");
        Serial.print(channel);
        Serial.print(" = ");
        Serial.println(value);
    }
    delay (1000);
}
```



Textbook 4051 connection

Problem

More analog sensors than available analog pins.

Solution

Use a 4051 to select one channel at a time.

Selector lines

S0, S1, S2 determine the selected channel.

How the 4051 idea works

- Several sensor signals are connected to separate channels.
- The multiplexer connects one selected channel to the common measurement path.
- Binary selector lines choose the active channel.
- The Arduino reads each channel sequentially and processes the values in software.

Measuring voltage up to 5 V

Vin ≈ ADC / 1023 × Vref

Measurement chain

input voltage → analog pin → analogRead() → calculation → Serial Monitor

Check first

Know the correct Vref before converting the code to volts.

Selector pins			Input channel
1	0	1	5
1	1	0	6
1	1	1	7

You may recognize the pattern in Table 5-3 as the binary representation of the decimal values from 0 to 7.

In the Solution sketch, `getValue()` is the function that sets the correct selector bits for the given channel using `digitalWrite(pin, isBitSet)` and reads the analog value from the selected 4051 input with `analogRead(analogPin)`. The code to produce the bit patterns uses the built-in `bitRead` function (see Recipe 3.12).

Bear in mind that this technique selects and monitors the eight inputs sequentially, so it requires more time between the readings on a given input compared to using `analogRead` directly. If you are reading eight inputs, it will take eight times longer for each input to be read. This may make this method unsuitable for inputs that change value quickly.

See Also

The Arduino Playground tutorial for the 4051

The 74HC4051 datasheet

Textbook measurement arrangement

Threshold-based response

- A measurement becomes useful when the program makes a decision from it.
- Compare the ADC value with thresholds or use a mapped range.
- Three regions can represent normal / warning / critical conditions.
- An LED may indicate the region, or the value may control another output.

Worked threshold example

Region 1

Vin < 2 V

Region 2

2 V ≤ Vin ≤ 4 V

Region 3

Vin > 4 V

$$2\ \text{V} \rightarrow \text{ADC} \approx 409$$

$$4\ \text{V} \rightarrow \text{ADC} \approx 818$$

Decision regions: <409 • 409–818 • >818

Why a voltage divider is needed

$$Vout = Vin \times R2 / (R1 + R2)$$

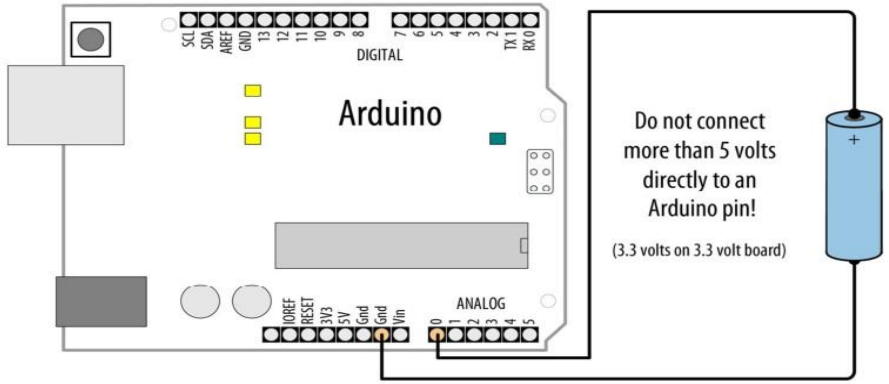
Example

Vin = 12 V, R1 = 30 kΩ, R2 = 10 kΩ → Vout = 3 V

Safety

Scale a higher external voltage before it reaches the analog pin.

Use **AnalogRead** to measure the voltage on an analog pin. Convert the reading to a voltage by using the ratio of the reading to the reference voltage (5 volts), as shown in Figure 5-9.



Measuring voltages up to 5 volts using 5V board

Measuring Analog Voltages on ESP8266 Boards

If you are using an ESP8266-based board, you may be limited to voltages in the range of 0 to 1 volt. Some ESP8266-based boards have built-in voltage dividers that allow you to read up to 3.3V (the ESP8266 itself runs at 3.3 volts), so be sure to check the documentation for your board. Without a voltage divider, the ESP8266 analog input pins max out at 1V. (See Recipe 5.11.)

Textbook voltage-divider example

Voltage-divider design thinking

- The divider is a ratio-forming network: the analog pin sees only a fraction of the external voltage.
- Choose resistor values so the maximum external voltage stays within the safe input range.
- Use the divider ratio again when reconstructing the original voltage in software.
- This is a good example of hardware + software working together.

Short program: digital input → LED output


```
pinMode(7, INPUT_PULLUP);
pinMode(13, OUTPUT);
int state = digitalRead(7);

if (state == LOW)
  digitalWrite(13, HIGH);
else
  digitalWrite(13, LOW);
```

Sequence

configure → read → interpret → act

Key logic

LOW means pressed because the pull-up is internal.

Exam

Write the complete short program, not isolated statements.

Dry run: analogRead() + map()


```
sensorValue = analogRead(A0);
value = map(sensorValue, 0, 1023, 0,
100);
Serial.println(value);
```

Given

analogRead(A0) = 512

Map

512 is roughly half of 0–1023 → about 50

Output

Serial Monitor prints about 50

Dry-run method: substitute → calculate → track variables → predict output.

Exam-style integrated problem

Exam-oriented review

An Arduino receives an analog signal and an LED must indicate three conditions: below 2 V, 2–4 V, and above 4 V. Analyse the decision structure, obtain the 10-bit ADC boundaries for a 5 V reference, and state the three decision regions.

Key points

- 2 V → about 409; 4 V → about 818.
- Regions: <409, 409–818, >818.
- The question combines analysis, calculation and decision logic.

CO / HHH mapping

CO3 • L4 (Head)

Write the answer compactly but include all requested elements.

Common analog-input mistakes

- Confusing an ADC code with a voltage value.
- Using the wrong Vref.
- Treating the ADC as continuous instead of discrete.
- Using map() as though it changes the physical voltage.
- Connecting an excessive voltage directly to the analog input.
- Forgetting to check wiring before blaming the code.

Analog-input practical bridge

BUILD

Potentiometer → analog pin
LED → output

MEASURE

analogRead()
Serial Monitor
ADC ↔ voltage

PROCESS

map()
thresholds
output response

The 2-hour practical should reinforce the same signal-processing chain used in the theory slides.

Chapter 5B recap

- A potentiometer or sensor creates a variable voltage.
- The ADC converts that voltage into a discrete digital code.
- analogRead() provides the code; Vref lets us convert it into volts.
- map() changes the numerical range for application logic.
- A 4051 can multiplex several analog channels onto one measurement path.
- Thresholds turn a measurement into a decision; a voltage divider scales higher external voltages safely.

2-Mark Question • Analog input

Exam-oriented review

Apply the ADC relationship for a 10-bit Arduino input with a 5 V reference to obtain the digital reading for 2.5 V. Also determine the approximate voltage represented by one ADC count.

Key points

- ADC ≈ (2.5/5)×1023 ≈ 512.
- One count ≈ 5/1023 ≈ 4.89 mV.
- Know the formula and the interpretation of the result.

CO / HHH mapping

CO2 • L3 (Head)

Write the answer compactly but include all requested elements.

2-Mark Question • 4051 multiplexer

Exam-oriented review

A system has more analog sensors than available Arduino analog input pins. Select an appropriate method using a 4051 multiplexer and indicate how its selector lines are used to access the individual sensor channels.

Key points

- Use a 4051 analog multiplexer.
- Selector lines determine which channel is connected.
- Read channels sequentially through the common path.

CO / HHH mapping

CO2 • L3 (Head)

Write the answer compactly but include all requested elements.

2-Mark Question • Voltage divider

Exam-oriented review

Determine the output voltage of a divider used to interface a 12 V source to an Arduino analog input when R1 = 30 kΩ and R2 = 10 kΩ. State why the divider is required in this application.

Key points

- Vout = 12×10/(30+10) = 3 V.
- The divider scales the source to a safer analog-input voltage.

CO / HHH mapping

CO2 • L3 (Head)

Write the answer compactly but include all requested elements.